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Article: Treatment

Effectiveness of medical nutrition treatment delivered by dietitians on glycaemic outcomes and lipid profiles of Arab, Omani patients with Type 2 diabetes

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Abstract

Aims In this randomized controlled trial we evaluated the effectiveness of medical nutritional therapy on Arab patients with Type 2 diabetes in Oman delivered by a dietitian.

Methods Patients with Type 2 diabetes (n = 170) were randomly assigned to a group receiving usual nutritional care (n = 85) or a group receiving practice guidelines nutritional care (n = 85). Anthropometric (weight, height, BMI and waist circumference) and biochemical (fasting blood glucose, HbA_{1c}) variables were measured at baseline and after each appointment. Patients were given 1–3 appointments with a dietitian over 6 months.

Results Those in the group receiving practice guidelines nutritional care (n = 85) had significant changes in HbA_{1c} (−0.8%, P = 0.001), fasting plasma glucose (−1.3 mmol\l, P = 0.003) and weight (−5.1 kg, P = 0.05), whereas the patients in the usual nutritional care group (n = 85) had no significant improvements in either HbA_{1c} (−0.4%, P = 0.248) or fasting plasma glucose (−0.2 mmol/l, P = 0.638) during the same period. We also found a significant difference between the group receiving practice guidelines nutritional care and the usual nutritional care group, respectively, in waist circumference (96.9 ± 7.9 vs. 100.0 ± 8.7 cm, P = 0.019), triglycerides levels (1.42 ± 0.58 vs. 1.98 ± 0.96 mmol\l, P = 0.001), cholesterol levels (5.1 ± 1.0 vs. 5.5 ± 0.9 mmol/l, P = 0.009) and LDL cholesterol levels (3.58 ± 0.98 vs. 3.89 ± 0.98 mmol/l, P = 0.046).

Conclusions Medical nutrition therapy provided by dietitians to Arab patients with Type 2 diabetes in Oman resulted in significant improvements in anthropometric and biochemical outcomes in both the usual nutritional care group and the group receiving practice guidelines nutritional care. Subjects with Type 2 diabetes tended to do better with practice guidelines nutritional care than with usual nutritional care. Ongoing medical counselling in nutrition by a trained dietitian is important for better long-term metabolic control.

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Keywords glycaemic outcomes, lipid profile, medical nutritional treatment, Oman patients with Type 2 diabetes

Introduction

Both developed and developing nations are facing an increasing incidence of diabetes mellitus. Diabetes and its complications are becoming a common cause of morbidity and mortality worldwide [1]. Studies have shown that diabetes and pre-diabetes are highly prevalent among the Arab population in the Middle East [2]. Intervention studies have demonstrated that diabetes could be reduced and prevented among high-risk individuals [3].

The first National Diabetes Survey in Oman was conducted in 1991. After approximately 10 years, a comparable nationwide survey was carried out to estimate non-communicable diseases and their risk factors during the first quarter of 2000, including Type 2 diabetes mellitus [4]. The prevalence of Type 2 diabetes has markedly increased in Oman over the past decade. More than 11% of the population aged ≥ 20 years was found to have diabetes compared with 8.3% in 1991. Thus, the burden of Type 2 diabetes has increased by one third over a single decade [2].

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Medical nutrition treatment is accepted to be the cornerstone of diabetes management. Present standards of medical care advocate individualized diet and nutritional recommendations, preferably by a registered dietitian familiar with the nutritional management of diabetes. The principles of medical nutrition treatment for subjects with diabetes include maintenance of near-normal blood glucose levels, achievement of optimum lipid levels and maintenance of a reasonable weight [5].

Practice guidelines for medical nutrition treatment provided by dietitians for patients with Type 2 diabetes have been developed [6]. They provide a framework to help the dietitian in the assessment, intervention and evaluation of outcomes of the medical nutrition treatment. Nutrition practice guidelines also integrate the medical nutritional therapy into systems of total care for the management of Type 2 diabetes. The primary objective was intention to treat. This study aims to investigate the effectiveness of practice guideline nutritional care compared with usual nutritional care on medical and clinical outcomes for patients with Type 2 diabetes in Oman.

Methods and research design

Subjects

Subjects were recruited at the outpatient diabetes clinic in Sultan Qaboos University in Muscat, Oman. The study was conducted during the period from April to September 2009. The Muslim fasting month of Ramadan was not included in the study period. Two hundred subjects, aged 30–70 years, with a definite diagnosis of Type 2 diabetes and who were free of chronic complications, were invited to participate in the study. This number of patients is estimated based on a power calculation to detect a 1.0 percentage point difference in the amount of change in HbA_{1c} levels between the group receiving practice guidelines nutritional care and the usual nutritional care group. At least 77 subjects were required in each group. To account for sample dropout during the period of the study, 23 subjects (approximately 25%) were added to each group. All patients gave informed consent and the study was approved by the medical research ethical committee of Sultan Qaboos University in Oman (MREC no. 286) and the Medical Research Ethics Committee, Faculty of Medicine and Health Sciences in University Putra Malaysia.

Inclusion and exclusion criteria

We recruited subjects for the study based on the following inclusion criteria. The subjects should be aged between 30 and 70 years and have been diagnosed as having Type 2 diabetes. They should be free of diabetic complications (such as gastroparesis and renal disease) and have consented to participate in the study. Subjects were excluded if they had had a recent heart attack or stroke (within the previous 6 months) or had had a serious illnesses (such as cancer or surgery within the previous 4 weeks). Patients were excluded from the study if they had illnesses requiring corticosteroid therapy or became pregnant, which might have an independent effect on outcomes.

Study design

Two hundred subjects were assigned randomly to the usual nutritional care group or the group receiving practice guidelines nutritional care. Each patient was given a number, which was put in a box. By random selection, the first 100 numbers were assigned to the group to receive practice guidelines nutritional care and the remaining numbers were assigned to the usual nutritional care group. Usual nutritional care consisted of one appointment of approximately 1 h with a dietitian, during which time data from the patient and the referring physician were used to develop a nutrition care plan. The practice guidelines nutritional care consisted of three appointments with a dietitian with specific assessments and intervention measures. Monk *et al.* developed practice guidelines nutritional care to be provided by dietitians for patients with Type 2 diabetes [6]. Practice guidelines for medical nutrition care recommend that subjects with Type 2 diabetes be referred to a dietitian within the first month after the diagnosis of Type 2 diabetes for a series of 2–3 appointments with the dietitian, approximately 2.5 h in total. On the basis of the referral data and goals, a suitable educational intervention and a nutrition prescription are planned and implemented. Short-term (4–6 weeks) behavioural goals (e.g. for diet, eating and exercise) that will lead to successful fulfillment of long-term goals are mutually identified. If the stated goals and the behavioural objectives for education are met, the need for extra information and a second follow-up appointment may be decided by the patient and the dietitian. If the goals and behavioural objectives have not been met, a second follow-up appointment should be scheduled. Both dietitian and the patient have the option of scheduling extra follow-up appointments. Ongoing self-management education is recommended at 6-month to 1-year intervals. Details of the practice guidelines nutritional care and the usual nutritional care appointments, assessment and intervention are shown in Figs 1 and 2.

Measurements and variables

All subjects were assessed at the time of enrollment and 3 and 6 months later. Anthropometric measurments, glucose levels and lipid profile of the patients were assessed. Anthropometric measurements included values for weight, height, BMI and waist circumference. The subject's height and weight were measured, with patients wearing light clothing and without shoes, by dietitians using an electric scale (Seca, Hamburg, Germany) or a beam-balance scale (Seca). A wall measure was used for height measurement. BMI was calculated as weight in kilograms divided by height in square metres (kg/m^2). Waist circumference was measured by dietitians using a cloth tape measure with subjects standing in a relaxed position. Waist circumference (umbilicus) was measured at the horizontal level that yielded the minimum measurement. All measurements were standardized to

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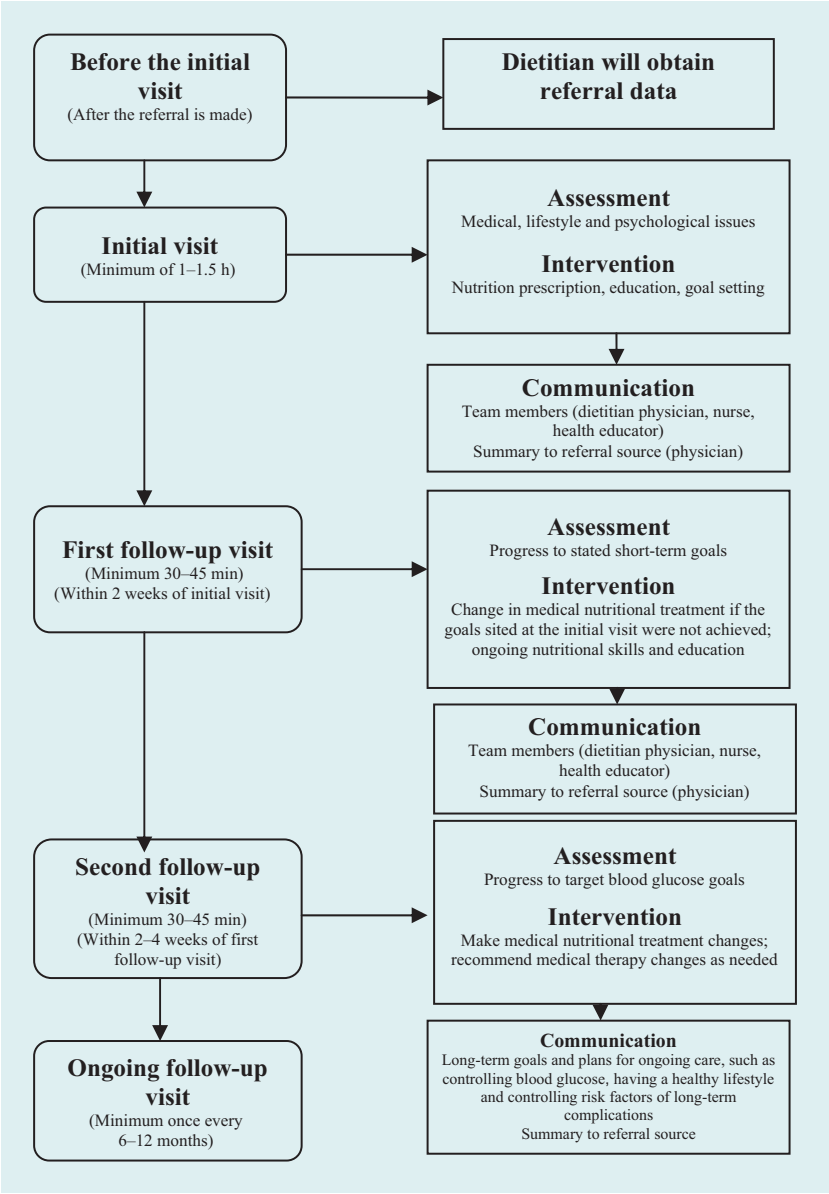

FIGURE 1 Practice guidelines nutritional care for patients with Type 2 diabetes.

ensure consistency. Weight loss of 1.5–3.0 kg at 4–6 weeks and a weight loss of 4.5–9 kg at 6 months were considered to be successful outcomes. BMI ≤ 25 kg/m^2 was considered to be a successful outcome.

At the time of entry to the study, at 3 months and at 6 months, patients with Type 2 diabetes came to the clinic after a 12-h fast. With the patient in a seated position, 30 ml of venous blood was drawn from the antecubital vein. Glycaemic control assessments included the assessment of fasting plasma glucose andHbA$_{1c}$ levels. Plasma glucose values were evaluated by the hexokinase method using the Baxter Paramax (Baxter Paramax, Irvine, CA, USA); HbA$_{1c}$ level was analysed by ion exchange high-performance liquid chromatography on a Bio-Rad Diamat analyzer (Bio-Rad Laboratories, Hercules, CA, USA). A fasting plasma glucose level of 4.4–7.8 mmol/l or a minimum 20% decrease from entry to 6 months was considered a successful outcome. An HbA$_{1c}$ value of 58 mmol/mol (7.5%) or less [normal measurement < 42 mmol/mol (6%)], or a minimum 20% decrease from initial appointment to 6-month appointment, was also considered a successful glucose outcome. Serum insulin level was not included as a variable in our present study because large numbers of the patients who participated have insulin injections as medication, which may interfere with insulin levels.

Lipid profile assessment includes the assessment of serum total cholesterol, LDL cholesterol, HDL cholesterol and serum triglycerides values. Values of lipids were determined by enzymatic methods on the Baxter Paramax. Total cholesterol

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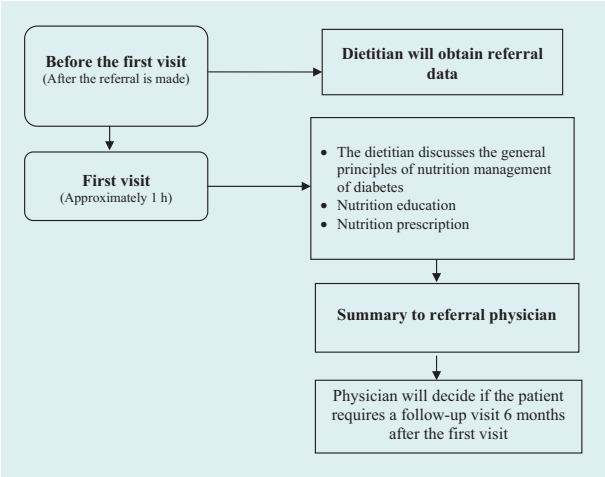

FIGURE 2 Usual nutritional care given to patients with Type 2 diabetes.

of ≤ 5.2 mmol/l, LDL cholesterol of ≤ 3.38 mmol/l and triglyceride levels of < 2.26 mmol/l were considered to be successful lipid outcomes. These levels reflect the recommendations of the internal local hospital standards of medical care for patients with diabetes mellitus.

Statistical analyses

Continuous variables between the two groups were analysed by ANOVA. The changes in variables between at entry, at 3 months and at 6 months in each group were analysed by paired *t*-test. The level for significance for all tests was $P < 0.05$. All of the analyses were carried out using the Statistical Package for Social Sciences, version 15.0 (SPSS Inc., Chicago, IL, USA).

Results

Baseline characteristics of the two groups

Out of the 200 patients with Type 2 diabetes who participated in this study, laboratory measurements at entry, 3 months and 6 months were complete for 170 subjects. The outcome analysis was based on the sample size of 170 subjects (85% of the original enrollees). No significant differences from entry data (age, gender, medical therapy) were noted among those who dropped out compared with those who completed the study.

The basic characteristics of the subjects in the two study groups are shown in Tables 1 and 2. The basic characteristics of the subjects in the two study groups did not differ significantly before intervention. Renal function and total blood count were within normal levels in both groups. At time of entry, 69.4% (n = 119) of all patients with Type 2 diabetes participating in the study were overweight. The differences between groups in weight and BMI were insignificant.

Changes of anthropometric and biomedical variables between the two intervention groups

Anthropometric variables

Compared with entry time, the mean weight values of the group receiving practice guidelines nutritional care decreased significantly at the 3-month point of the study and these levels were sustained at 6 months. Examination using the diabetes

Table 1 Characteristics of the study subjects in the usual nutritional care (UNC) group and the group receiving practice guidelines nutritional care (PGC) before the intervention

	UNC group (n = 85)	PGC group (n = 85)	All (n = 170)	
Variables	Mean ± SD	Mean ± SD	Mean ± SD	P
Age (years)	52.0 ± 9.2	49.5 ± 11.3	50.7 ± 10.4	0.122
Diabetes duration (years)	4.1 ± 2.3	4.3 ± 2.1	4.2 ± 2.3	0.542
Weight (kg)	72.2 ± 12.9	73.3 ± 13.6	72.7 ± 13.0	0.586
Body mass index (kg/m2)	28.4 ± 5.2	27.6 ± 5.3	28.0 ± 5.2	0.313
Waist circumference (cm)	97.9 ± 8.3	97.2 ± 7.5	97.5 ± 7.9	0.554
Fasting plasma glucose (mmol/l)	12.1 ± 3.3	12.4 ± 3.1	12.3 ± 3.2	0.466
HbA_{1c}				
(mmol/mol)	88	89	89	0.632
(%)	10.2 ± 2.4	10.3 ± 1.8	10.3 ± 2.1	
Cholesterol (mmol/l)	5.4 ± 0.9	5.3 ± 1.0	5.4 ± 1.0	0.825
LDL cholesterol (mmol/l)	3.46 ± 0.88	3.57 ± 0.93	3.52 ± 0.90	0.437
HDL cholesterol (mmol/l)	1.08 ± 0.22	1.05 ± 0.23	1.07 ± 0.23	0.345
Triglycerides (mmol/l)	1.80 ± 0.87	1.64 ± 0.70	1.72 ± 0.79	0.187
Urea (mmol/l)	5.3 ± 1.4	5.6 ± 1.2	5.5 ± 1.3	0.084
Creatinine (μmol/l)	62.8 ± 17.0	62.4 ± 12.8	62.6 ± 15.0	0.839
White blood cells (10^9/L)	6.2 ± 2.1	6.0 ± 1.5	6.1 ± 1.8	0.459
Haemoglobin (g/dl)	12.5 ± 0.86	12.3 ± 1.4	12.4 ± 1.1	0.224
Platelets (10^9/L)	280 ± 68	291 ± 61	286 ± 65	0.309

Data are expressed as mean ± SD.

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Table 2 Gender, medical nutritional treatment (MNT) and education for the study subjects in the usual nutritional care (UNC) group and the group receiving practice guidelines nutritional care (PGC)

Variables	UNC group	PGC group	All
Gender			
Male	*n* = 33 (38.8%)	*n* = 38 (44.7%)	*n* = 71 (41.8%)
Female	*n* = 52 (61.2%)	*n* = 47 (55.3%)	*n* = 99 (58.2%)
Diabetes duration			
Less than 4 years	*n* = 10 (11.8%)	*n* = 22 (25.9%)	*n* = 32 (37.6%)
More than 4 years	*n* = 75 (88.2%)	*n* = 63 (74.1%)	*n* = 53 (62.4%)
MNT			
MNT alone	*n* = 11 (13%)	*n* = 10 (11.8%)	*n* = 21 (12.4%)
MNT and oral agents	*n* = 37 (43.5%)	*n* = 45 (52.9%)	*n* = 82 (48.2%)
MNT and combination therapy (oral agents and insulin)	*n* = 37 (43.5%)	*n* = 30 (35.3%)	*n* = 67 (39.4%)
Education			
Less than high school	*n* = 50 (58.8%)	*n* = 48 (56.5%)	*n* = 98 (57.6%)
High school	*n* = 15 (17.6%)	*n* = 17 (20%)	*n* = 32 (18.8%)
College	*n* = 20 (23.5%)	*n* = 20 (23.5%)	*n* = 40 (23.5%)

Data are expressed as total number (*n*) or percentage (%).

therapy regime indicated that the group treated with medical nutritional treatment alone and the group treated with medical nutritional treatment and oral hypoglycaemic agents experienced a greater weight loss than the other therapy groups (from a baseline weight of 73.4 ± 13.8 to 68.2 ± 13.2 kg at 6 months). The group receiving medical nutritional treatment and a combination therapy of insulin and oral agents also experienced a weight loss from baseline (73.1 ± 13.5 to 68.4 ± 14.5 kg) at 6 months. Weight changes were reflected in similar changes in BMI. At 6 months, 55.3% (*n* = 47) of the group receiving practice guidelines nutritional care had achieved the target outcome (≥ 4.5 kg weight loss).

In the usual nutritional care group, a weight loss from baseline to 6 months was insignificant. This trend was reflected in the BMI of the usual nutritional care group. The same trend was observed between diabetes treatment regimes. Patients in the medical nutritional treatment alone group, or those receiving medical nutritional treatment with oral agents, or medical nutritional treatment with combination therapy treated by basic nutritional treatment show insignificant changes in weight and BMI. At 6 months, 14% (*n* = 12) of the UNC group had achieved a successful weight loss of 4.5 kg or more. Mean total waist circumference did not change significantly.

The group receiving practice guidelines nutritional care and the usual nutritional care group did not show any significant difference between their weight and BMI values during the study period. In contrast, the waist circumference measurements of the group of patients receiving practice guidelines nutritional care show a significant decrease compared with patients in the usual nutritional care group at the 6-month point, as shown in Table 2.

Glycaemic control

Analyses of data in the group receiving practice guidelines nutritional care (*n* = 85) show that significant improvements in both fasting plasma glucose and HbA_{1c} levels were found when each value was compared with the value at entry (Table 3).

The reduction was greatest between entry and 3 months. During the 3- to 6-month point, both fasting plasma glucose and HbA_{1c} levels began to show a slight increase; however, the changes were still significantly lower at 6 months compared with entry values. In the usual nutritional care group, HbA_{1c} values insignificantly decreased at 3 months compared with values at time of entry to the study, and then an upward trend was shown at 6 months. The same trend of an insignificant reduction during the first 3 months of the study and then a slight increase in the levels of fasting plasma glucose was observed in the usual nutritional care group. Changes in glycaemic variables are shown in Figs 3 and 4.

Approximately 57% (*n* = 97) of all participants (*n* = 170) achieved their target goal of a fasting plasma glucose level of ≤ 7.8 mmol/l and HbA_{1c} values of ≤ 58 mmol/mol (7.5%) — or a 20% decrease after 6 months. At 3 months, participants assigned to the group receiving practice guidelines nutritional care had reduced their entry fasting plasma glucose mean levels by 16.7% and their HbA_{1c} mean levels by 13.1%. The usual nutritional care group demonstrated a mean 7.3% reduction in fasting plasma glucose level and an 8% reduction of HbA_{1c} level at the 3-month point. At 6 months, those in the group receiving practice guidelines nutritional care had reduced their mean fasting plasma glucose level by 11.2% and their HbA_{1c} level by 8.2% compared with levels at entry to the study. Figures 1 and 2 show changes in fasting plasma glucose and HbA_{1c} levels in the group receiving practice guidelines nutritional care and the usual nutritional care group.

As shown in Figs 3 and 4, the group receiving practice guidelines nutritional care had a greater improvement in both indicators of glycaemic control during the 6-month point. The difference between fasting plasma glucose values in the two groups was significant after 3 months of study. In contrast, the

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Table 3 Means and standard deviations for metabolic and clinical outcomes at entry, 3 months and 6 months for practice guidelines nutritional care (PGC) and usual nutritional care (UNC) provided by dietitians for subjects with Type 2 diabetes mellitus

	PGC (n = 85)			UNC (n = 85)		
Variables	Entry Mean ± SD	3 months Mean ± SD	6 months Mean ± SD	Entry Mean ± SD	3 months Mean ± SD	6 months Mean ± SD
Weight (kg)	73.3 ± 13.6	68.7 ± 13.9 0 vs. 3 months*	68.2 ± 13.4 0 vs. 6 months*	72.2 ± 12.9	69.7 ± 12.2	69.6 ± 12.1
BMI (kg/m^2)	27.6 ± 5.3	26.3 ± 5.7 0 vs. 3 months*	26.2 ± 5.3	28.4 ± 5.2	27.4 ± 4.9	27.4 ± 5.0
Waist circumference (cm)	97.2 ± 7.5	96.9 ± 7.6	96.9 ± 7.9‡	97.9 ± 8.3	99.0 ± 8.7	100.0 ± 8.7
Fasting plasma glucose (mmol/l)	12.4 ± 3.2	10.4 ± 2.5† 0 vs. 3 months***	11.1 ± 2.8 0 vs. 6 months**	12.1 ± 3.3	11.2 ± 3.0	11.9 ± 3.2
HbA_{1c}						
(mmol/mol)	89	75	80	88	79	84
(%)	10.3 ± 1.8	9.0 ± 1.8 0 vs. 3 months***	9.5±1.4[b] 0 vs. 6 months** 3 vs. 6 months*	10.2 ± 2.4	9.4 ± 2.0	9.8 ± 1.8
Cholesterol (mmol/l)	5.3 ± 1.0	5.1 ± 1.0	5.1 ± 0.97[b]	5.4 ± 0.9	5.1 ± 0.8 0 vs. 3 months*	5.5 ± 0.9 3 vs. 6 months**
LDL cholesterol (mmol/l)	3.57 ± 0.93	3.74 ± 1.01	3.58 ± 0.98[b]	3.37 ± 0.88	3.63 ± 0.75	3.66 ± 0.87
HDL cholesterol (mmol/l)	1.05 ± 0.23	1.01 ± 0.33	1.09 ± 0.32	1.08 ± 0.22	1.05 ± 0.25	1.01 ± 0.22 0 vs. 6 months*
Triglycerides (mmol/l)	1.64 ± 0.70	1.39 ± 0.62[a] 0 vs. 3 months*	1.42 ± 0.58‡ 0 vs. 6 months*	1.80 ± 0.87	1.70 ± 0.75	1.98 ± 0.96 3 vs. 6 months*

Data were expressed as mean ± SD.
*P < 0.05, **P < 0.01, ***P < 0.001.
†P = significant difference between two groups at 3-month point.
‡P = significant difference between two groups at 6-month point.

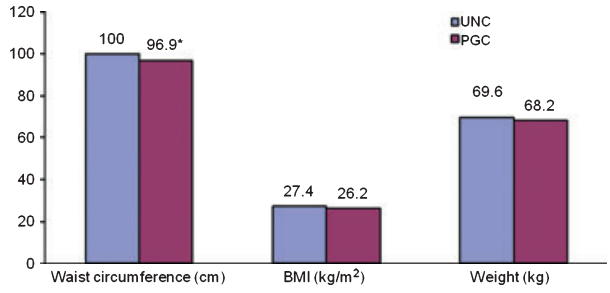

FIGURE 3 Weight, BMI and waist circumference of the group receiving practice guidelines nutritional care (PGC) and the usual nutritional care (UNC) group at the 6-month point of the study. *P < 0.05.

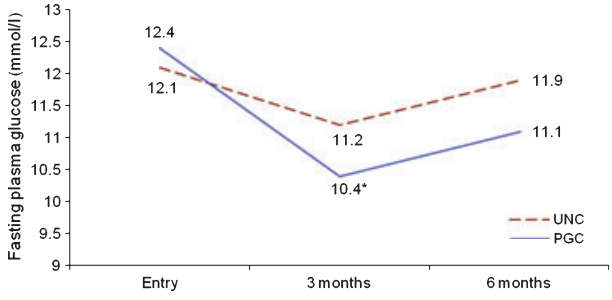

FIGURE 4 Changes in fasting plasma glucose level in the group receiving practice guidelines nutritional care (PGC) and usual nutritional care (UNC) group. *P < 0.05.

difference in HbA_{1c} values between the group receiving practice guidelines nutritional care and the usual nutritional care groups at the 3-month point did not reach statistical significance. HbA_{1c} values of both groups continued their reduction trend, but still there was no significant difference observed between mean HbA_{1c} levels of the two groups at the end of the 6-month point.

Lipid profiles

Changes in the values of total cholesterol, LDL cholesterol, HDL cholesterol and triglycerides in the group receiving practice guidelines nutritional care are shown in Table 3. In that group, total cholesterol and LDL cholesterol levels dropped insignificantly at 3 months. LDL cholesterol levels began an upward trend to 6 months, whereas total cholesterol stayed at the same levels. At the end of 6 months of the study, levels of HDL cholesterol showed an insignificant increase compared with levels at time of time. At 6 months, 51.8% (n = 44) of the group receiving practice guidelines nutritional care had achieved or maintained a target cholesterol outcome (≤ 5.2 mmol/l), 42.4% (n = 36) a target LDL cholesterol outcome (≤ 3.38 mmol/l) and 94.1% (n = 80) a target triglyceride outcome (≤ 2.26 mmol/l).

Mean total cholesterol levels in the usual nutritional care group showed a significant change at 3 months, with an upward trend at 6 months. There was no significant change in the usual nutritional care group levels of LDL cholesterol and triglycerides between time of entry and after 6 months of study. The HDL

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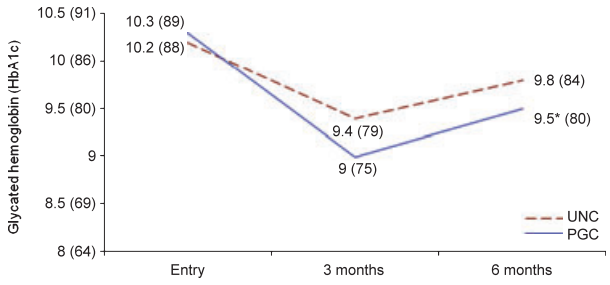

FIGURE 5 Changes in HbA_{1c} level in the group receiving practice guidelines nutritional care (PGC) and usual nutritional care (UNC) group. $^*P < 0.05$.

cholesterol mean value at 6 months was significantly lower than values at time of study entry.

As shown in Fig. 5, those in the group receiving practice guidelines nutritional care have better total triglyceride levels at the 3-month point of the study compared with the usual nutritional care group. Lipid profiles of the group receiving practice guidelines nutritional care tend to be enhanced significantly after 6 months of the intervention. Figure 6 show the changes in cholesterol, LDL-cholesterol, HDL-cholesterol and triglyceride levels in PGC group and UNC group at 6 months of study. Cholesterol, LDL cholesterol and triglycerides values of the patients in the group receiving practice guidelines nutritional care were significantly lower than the values of the patients receiving usual nutritional care.

Changes in therapy

In both study groups, changes in type of diabetes treatment during the 6 months of the study were recorded. In the total study group (n = 170) at time of entry, 12.4% (n = 21) of subjects were treated with medical nutritional treatment alone, 48.2% (n = 82) were treated with medical nutritional treatment and oral hypoglycaemic agents and 39.4% (n = 67) were treated with medical nutritional treatment and a combination of oral agents and insulin therapy. At the 6-month point, 18.2% (n = 31) of the total study group were treated by medical nutritional treatment alone, 47.1% (n = 80) were treated by medical nutritional treatment and oral agent, and 34.7% (n = 59) by medical nutritional treatment with a combination therapy of insulin and oral agents.

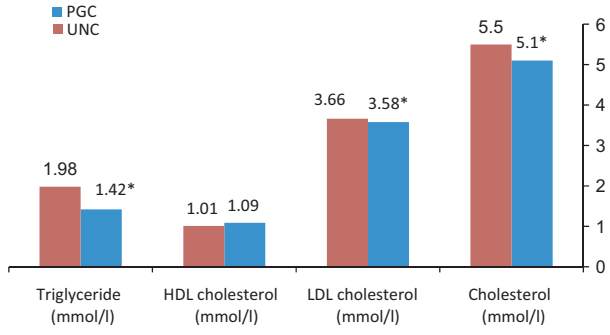

FIGURE 6 Changes in cholesterol, LDL cholesterol, HDL cholesterol and triglyceride levels in the group receiving practice guidelines nutritional care (PGC) and the usual nutritional care (UNC) group at the 6-month point of the study. $^*P < 0.05$.

In the total group (n = 170), 20 changes in therapy (12% of the subjects) were reported. Of these 20 changes, six occurred in the usual nutritional care group (7% of the subjects) and 14 occurred in the group receiving practice guidelines nutritional care (16.5% of the patients). All 14 of the changes in the group receiving practice guidelines nutritional care were recommended by the dietitian to the treating physician and all of these changes were implemented.

Discussion

Dietary intervention delivered by a dietitian resulted in significant improvements in biochemical and anthropometric variables of the subjects in the study. This study showed several biochemical and anthropometric changes. Weight loss was one of the determinants of biochemical improvements of our patients. The weight loss achieved in our study is a successful outcome for patients with Type 2 diabetes. This suggests that subjects who lose 4.5 kg weight, and who meet physical activity and dietary fat goals, could reduce their risk of diabetes and its complications. These estimates are similar to the findings of the Finnish Diabetes Prevention Study [3].

In our study, the loss of weight in the group receiving practice guidelines nutritional care was significant ($P < 0.05$), with similar significant reductions seen in fasting plasma glucose ($P < 0.05$) and HbA_{1c} ($P < 0.05$) levels.

Recent clinical trials [7,8] have established the feasibility of interrupting the worsening of hyperglycaemia in overweight people with diabetes. In 2006, in the USA, the Diabetes Prevention Program Research Group found that lifestyle interventions succeeded in producing long-term weight loss and that weight loss was associated with prevention of diabetes [9].

Medical nutrition treatment provided by dietitians in the form of dietary assessment, intervention and evaluation of outcomes resulted in significant biochemical improvements in our studied subjects. This was evidenced by significant improvements in glycaemic control represented by fasting plasma glucose and HbA_{1c} levels and serum triglyceride level. Enhancement in glycaemic control was greater for patients in the group receiving practice guidelines nutritional care, although the difference between that group and the usual nutritional care group in fasting plasma glucose levels did not reach statistical significance.

The most significant changes occurred from the time of study entry to 6 weeks, and improvement continued to the 3-month point. After 3 months, in both groups, an upward trend was noted in glucose values, which suggests a need for additional interventions by dietitians to prevent this upward shift. This trend suggests a need for more studies documenting the long-term effectiveness of medical nutritional therapy.

More intensive medical nutritional treatment in accord with practice guidelines nutritional care resulted in significantly better

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glycaemic values compared with the administration of usual nutritional care in Arab subjects with Type 2 diabetes mellitus. Similar results were found in a randomized controlled trial in Taiwan [10], which evaluated the effect of registered dietitian-led management of diabetes on glycaemic control and macronutrient intake in patients with Type 2 diabetes in primary care clinics. The authors of that study found that patients with Type 2 diabetes with poorly controlled baseline HbA_{1c} [≥ 53 mmol/mol (7%)] had significantly greater improvements in HbA_{1c} and fasting plasma glucose than the control subjects over the period of 1 year. They concluded that dietitian-led management of diabetes can improve glycaemic control in patients with poor glycaemic control.

The primary goal of the intervention in this study was to improve glycaemic control, but lipid levels are also important in the management of Type 2 diabetes mellitus. The improvement in lipid levels was greater for patients in the group receiving practice guidelines nutritional care, with a significant difference in serum triglyceride, total cholesterol and LDL cholesterol values.

Of the study subjects, 32 (18.8%) had triglyceride levels greater than 2.27 mmol/l and 38 (22.4%) had HDL cholesterol levels less than 0.9 mmol/l at the time of entry to the study. Milagros *et al*. found that approximately 50% of persons with diabetes have elevated plasma triglyceride levels (≥ 1.7 mmol/l) and low HDL cholesterol levels (≤ 1.0 mmol/l) (12). Our study used a lower cut-off point for triglyceride levels and a higher HDL cholesterol level; our results are therefore almost certainly similar to reported values. Comparable results were found in another study in India by Rakesh *et al*., who reported that dyslipidaemia with low HDL cholesterol was the most common pattern among men and women, contributing to 33% of patients with Type 2 diabetic dyslipidaemia [13].

In our study, approximately 100 (59%) of the patients with Type 2 diabetes had initial cholesterol levels above 5.2 mmol/l, and 72 (42.4%) had an LDL cholesterol level > 3.4 mmol/l. These levels are similar to those that have been reported for patients with diabetes [12].

We attribute the achieved improvements seen in our subjects to the continuous counselling delivered by dietitians, thus showing the importance of the role of the dietitian in diabetes care. Wynn *et al*. investigated the management of patients with nutrition-related issues and whether implementation of nutrition counselling guidelines is feasible in primary care practices. They found that, among the 757 physicians surveyed, respondents had positive attitudes about the role of nutrition in patient health, and most physicians (58.1%) believed that more than 60% of their patients would benefit from nutrition counselling [15].

Study limitations

There are some limitations in the current study. The education level of the subjects in our study tended to be of a lower class, with less than a high school education, so the results may not apply to all persons with Type 2 diabetes. Twenty-four-hour dietary recall is used in nutritional care in both interventions to estimate the food intake of subjects with diabetes. However, 24-h dietary recall could be biased and interviews of patients by trained dietitians who are able to retrieve more accurate dietary information may attenuate such errors. Furthermore, we did not analyse serum insulin levels, therefore the present study did not report changes in serum insulin level.

Conclusion

Medical nutritional therapy provided by dietitians to Arab patients with Type 2 diabetes in Oman resulted in significant improvements in anthropometric and biochemical outcomes in both the usual nutritional care group and the group receiving practice guidelines nutritional care. Subjects with Type 2 diabetes tended to do better with practice guidelines nutritional care than with usual nutritional care. Because of the increased trend in glucose values after 3 months, continuous medical nutrition counselling by a trained dietitian is important for better long-term metabolic control. Systems of care that provided planned, integrated, consistent and continuous care from an experienced dietitian are important in the management of Type 2 diabetes. Our study clearly demonstrates the desirability and effectiveness of making dietitians full participants in the multidisciplinary team for diabetic care.

Competing interests

Nothing to declare.

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